Carbon Emission Dataset: Report Summary

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Dataset: `carbon_emission_dataset_with_Industry.csv`

1. Introduction

This report summarizes a complete Python analysis of the **Carbon Emission Forecasting Dataset**. The assignment document requires students to use Pandas, NumPy, Seaborn, and Matplotlib to clean, explore, visualize, and interpret a real-world carbon-emission dataset. The project description identifies the target variable as `Carbon_Emission_tCO2e_TARGET`, which represents carbon emissions measured in tonnes of carbon dioxide equivalent. The public Kaggle dataset is described as daily company-level operational, financial, energy, logistics, and ESG-related records designed for carbon-footprint accounting and emission prediction.1

> According to the project description, each record represents a daily company observation integrating energy usage, supply-chain activity, production output, industry sectors, and environmental indicators. The target variable is carbon emission in tCO2e.2

The downloaded CSV contains **18,250 original records** and **22 original columns**. After cleaning and feature engineering, the analysis dataset contains **18,250 records** and **28 columns**. The additional columns support interpretation of renewable share, non-renewable share, emission intensity, and time-based patterns.

2. Data Cleaning and Preparation

The data-cleaning workflow converted the `Date` column to datetime format, removed exact duplicate records, checked missing values, and preserved the original numerical and categorical fields. No missing values were found in the original dataset columns, and no duplicate records remained after cleaning. Additional variables were created to support deeper analysis: **actual renewable share**, **actual non-renewable share**, **emission intensity per production unit**, **month**, **year-month**, and **day of week**.

Cleaning Step	Result
Original records	18,250
Missing values after cleaning	0
Duplicate records after cleaning	0
Cleaned/engineered columns	28
Date range	2024 daily records

3. Required Descriptive Statistics

The assignment requires the mean, median, minimum, and maximum for five operational variables. Standard deviation is also included because it is useful for comparing variability across emission-related sources.

	mean	median	min	max	std
Total_Energy_Consumption_kWh	124777	124729	50008.7	199989	43106.6
Renewable_Energy_Consumption_kWh	43759.1	38888	5041.2	119005	24291.7
NonRenewable_Energy_Consumption_kWh	81017.9	77007.8	20483.3	179431	33957.9
Production_Output_Units	5538.96	5556.4	1000.47	9999.78	2597.58
Supply_Chain_Transport_km	2734.81	2735.02	500.38	4999.89	1300.15

The target variable has a mean of **32.365 tCO2e**, a median of **30.155 tCO2e**, a minimum of **6.815 tCO2e**, and a maximum of **86.572 tCO2e**. This indicates a wide range of emission intensities among company-day records.

Figure 1. Distributions of the five required operational variables

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4. Exploratory Data Analysis

The categorical frequency plots show the representation of sector, industry sector, transport mode, and carbon-reduction strategy. The dataset is balanced across transport modes and nearly balanced across the four industry-sector labels, while company records appear across five broader sectors.

Figure 2. Frequency bars for key categorical variables

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The line plot of average energy consumption over time shows that total energy use is consistently higher than each of its renewable and non-renewable components. Non-renewable energy is generally the larger component, which is important because non-renewable energy is also the variable most strongly associated with emissions.

Figure 3. Average total, renewable, and non-renewable energy over time

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The project PDF requests a scatter plot of emissions against temperature. However, the actual CSV does **not** include a temperature column. Therefore, the notebook and scripts explicitly check for any temperature-like field and report that the requested temperature analysis cannot be produced from this provided dataset. The available emissions-versus-time scatter plot is included below.

Figure 4. Carbon emissions versus time

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5. Bivariate and Multivariate Analysis

The strongest correlations with the target variable are concentrated in the energy and carbon-tax variables. In particular, **non-renewable energy consumption** has the strongest positive relationship with emissions, followed by carbon tax and total energy consumption.

	Correlation
NonRenewable_Energy_Consumption_kWh	0.9372
Carbon_Tax_USD	0.8557
Total_Energy_Consumption_kWh	0.7778
Energy_Cost_USD	0.6262
NonRenewable_Share_Actual_Percent	0.5006
Renewable_Share_Actual_Percent	-0.5006
Emission_Intensity_tCO2e_per_Unit	0.4886
Expected_Renewable_Share_Percent	-0.4768
Renewable_Energy_Consumption_kWh	0.07
Expected_Carbon_Reduction_Percent	0.0103

The following variable pairs meet the threshold of absolute correlation greater than or equal to **0.70**.

variable_1	variable_2	correlation
Total_Energy_Consumption_kWh	NonRenewable_Energy_Consumption_kWh	0.827
Total_Energy_Consumption_kWh	Carbon_Emission_tCO2e_TARGET	0.7778
Total_Energy_Consumption_kWh	Energy_Cost_USD	0.8071
Renewable_Energy_Consumption_kWh	Expected_Renewable_Share_Percent	0.7082
Renewable_Energy_Consumption_kWh	Renewable_Share_Actual_Percent	0.7382
Renewable_Energy_Consumption_kWh	NonRenewable_Share_Actual_Percent	-0.7382
NonRenewable_Energy_Consumption_kWh	Carbon_Emission_tCO2e_TARGET	0.9372
NonRenewable_Energy_Consumption_kWh	Carbon_Tax_USD	0.8028
Carbon_Emission_tCO2e_TARGET	Carbon_Tax_USD	0.8557
Expected_Renewable_Share_Percent	Renewable_Share_Actual_Percent	0.9582
Expected_Renewable_Share_Percent	NonRenewable_Share_Actual_Percent	-0.9582
Renewable_Share_Actual_Percent	NonRenewable_Share_Actual_Percent	-1

Figure 5. Correlation heatmap of numerical variables

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The boxplot grouped by `Industry_Sectors` shows that all four industry-sector categories have broadly similar emission distributions. Automotive Industry has the highest mean emissions, but the difference is small compared with the full within-sector spread.

Figure 6. Carbon emissions by industry sector

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The selected-feature pairplot confirms the same pattern visible in the correlation table: non-renewable energy consumption and total energy consumption move most clearly with the carbon-emission target, while production output and supply-chain transport distance show weaker direct patterns.

Figure 7. Pairplot of selected operational and emission variables

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6. Group-Based Analysis

The available categorical variables permit group-based analysis by broad `Sector` and by `Industry_Sectors`. The sector with the highest mean emissions is **IT**, while the industry-sector category with the highest mean emissions is **Automotive Industry**. These differences are numerically small, so the more important explanatory relationships are the energy and carbon-tax variables rather than sector labels alone.

Emissions by Sector

	count	mean	median	std	min	max
Agriculture	4380	32.41	30.298	14.532	7.638	83.336
Energy	3650	32.167	29.905	14.781	7.552	86.572
IT	1460	32.496	30.508	14.011	7.153	85.678
Manufacturing	5110	32.45	30.291	14.29	6.944	84.411
Transport	3650	32.338	29.887	14.579	6.815	84.978

Emissions by Industry Sector

	count	mean	median	std	min	max
Automotive Industry	4562	32.404	30.304	14.44	6.903	86.572
Cement Production	4563	32.361	30.02	14.324	7.031	85.678
Logistics	4562	32.353	30.347	14.484	6.815	82.279
Steel Manufacturing	4563	32.342	29.981	14.683	6.945	84.616

Figure 8. Average emissions by sector and industry sector

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7. Time-Series Exploration

The time-series workflow aggregated observations by `Date` and calculated daily average emissions. Moving averages were computed using **7-day** and **30-day** windows, while rolling variance was computed using a **30-day** window. The fitted linear trend slope is **0.000101 tCO2e per day**, indicating a modest **upward** movement over the year. The first 30-day average was **32.408 tCO2e**, while the last 30-day average was **32.896 tCO2e**, representing a **1.508%** increase.

Figure 9. Emissions time series with moving averages

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The rolling variance plot shows how short-term volatility changes through the year. Because the data are daily averages across 50 companies, the series is smoother than individual company-level emissions would be.

Figure 10. Thirty-day rolling variance of average daily emissions

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The additive decomposition used a **30-day period**. The estimated seasonal strength is **0.1001**, which indicates a **weak seasonal component**. Since the dataset covers one calendar year, seasonal interpretations should be treated as exploratory rather than definitive.

Figure 11. Additive decomposition of daily average emissions

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8. Answers to the Required Analytical Questions

Q1: Which emission sources have the highest variability?
The coefficient of variation was used because the variables have different measurement scales. The highest relative variability is observed in **Renewable_Energy_Consumption_kWh**.

	std	mean	coefficient_of_variation
Renewable_Energy_Consumption_kWh	24291.7	43759.1	0.5551
Supply_Chain_Transport_km	1300.15	2734.81	0.4754
Raw_Material_Usage_kg	20110.5	44869.4	0.4482
NonRenewable_Energy_Consumption_kWh	33957.9	81017.9	0.4191
Total_Energy_Consumption_kWh	43106.6	124777	0.3455

Q2: Are emissions normally distributed?
No. The target distribution has skewness of **0.6664** and kurtosis of **-0.0037**. The D'Agostino-Pearson normality test gives a p-value of **0.0**, so the analysis **reject normality at alpha=0.05**.

Q3: Which variables are strongly correlated?
The strongest target-related correlations are with **NonRenewable_Energy_Consumption_kWh** (**0.9372**), **Carbon_Tax_USD** (**0.8557**), and **Total_Energy_Consumption_kWh** (**0.7778**). These findings suggest that fossil/non-renewable energy use is the dominant emission driver in this dataset.

Q4: Is there a trend in emissions?
Yes, there is a modest upward trend. The slope is **0.000101 tCO2e per day**, and the last 30-day average is **1.508%** higher than the first 30-day average.

Q5: Do seasonal patterns exist?
The decomposition detects only a weak seasonal component, with seasonal strength **0.1001**. Seasonal

patterns are visible in the decomposition plot, but they are not strong and should be interpreted cautiously because the sample covers one year.

9. Conclusion

The analysis satisfies the assignment requirements by cleaning the data, generating descriptive statistics, producing EDA charts, examining bivariate and multivariate relationships, creating a correlation heatmap and pairplot, building boxplots, and performing time-series analysis with moving averages, rolling variance, and decomposition. The main conclusion is that **non-renewable energy consumption is the strongest emission-related driver** in the dataset. Carbon tax is also strongly correlated with the emission target, which is expected because tax values are typically linked to emission volume. Sector and industry-sector differences exist but are small compared with variation caused by energy-use variables.

References